

Assignment 2

Numerical Linear Algebra (NLA)

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21 Oct. 2025

Exercise 1

Write a nonrecursive algorithm in the spirit of row-oriented forward substitution that implements column-oriented forward substitution. Use a single array that contains $\mathbf{b}$ initially, stores intermediate results (e.g. $\hat{b}, \tilde{b}$) during the computation, and contains $\mathbf{y}$ at the end. Use your algorithm to write a Julia program that solves the system $\mathbf{Ax} = \mathbf{b}$ where

$$\mathbf{A} = \begin{pmatrix} 5 & 0 & 0 \\ 2 & -4 & 0 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -15 \\ -2 \\ 10 \end{pmatrix}$$

Exercise 2

Consider a system of n carts connected by springs, as shown in Figure 1. The i th spring has a stiffness of k_i newtons/meter. Suppose that the carts are subjected to steady forces of $f_1, f_2, \dots, f_n$ newtons, respectively, causing displacements of $x_1, x_2, \dots, x_n$ meters, respectively.

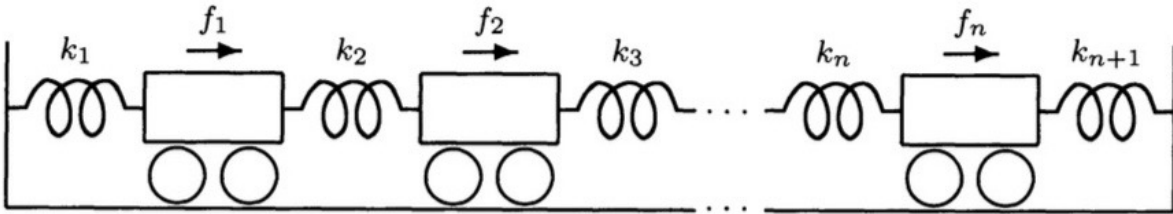


Figure 1: Mass-spring system

(a) Write down a system of n linear equations $\mathbf{Ax} = \mathbf{b}$ that could be solved for $x_1, \dots, x_n$. Since all nonzeros are confined to three diagonals, $\mathbf{A}$ is tridiagonal.

(b) Compute the solution in the case $n = 20$, $k_i = 1$ for all i , and $f_i = 0$ except $f_5 = 1$, $f_{16} = -1$. (Install ‘ToeplitzMatrices.jl’ in Julia to define $\mathbf{A}$ easily.)

Exercise 3

(a) The conductance C of a resistor is $C = 1/R$. Show that if two nodes with voltages x_i and x_j are connected by a resistor with conductance C , then the power drawn is:

$$\Lambda = \begin{bmatrix} x_i & x_j \end{bmatrix} \begin{bmatrix} C & -C \\ -C & C \end{bmatrix} \begin{bmatrix} x_i \\ x_j \end{bmatrix}$$

When is $\Lambda > 0$?

(b) Consider the circuit in Figure 2. Show that the power drawn is a sum of terms of the form $C(x_i - x_j)^2$, and is positive unless all $x_1 = x_2 = \dots = x_6$.

(c) Show that the power can be written as $x^T H x$, where H is a symmetric 6×6 matrix. Show $x^T H x \geq 0$ for all x , so H is positive semi-definite.

(d) Derive the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ for the circuit. Show that $\mathbf{A}$ is a submatrix of H and is positive definite. Use Julia to compute a Cholesky factor of $\mathbf{A}$.

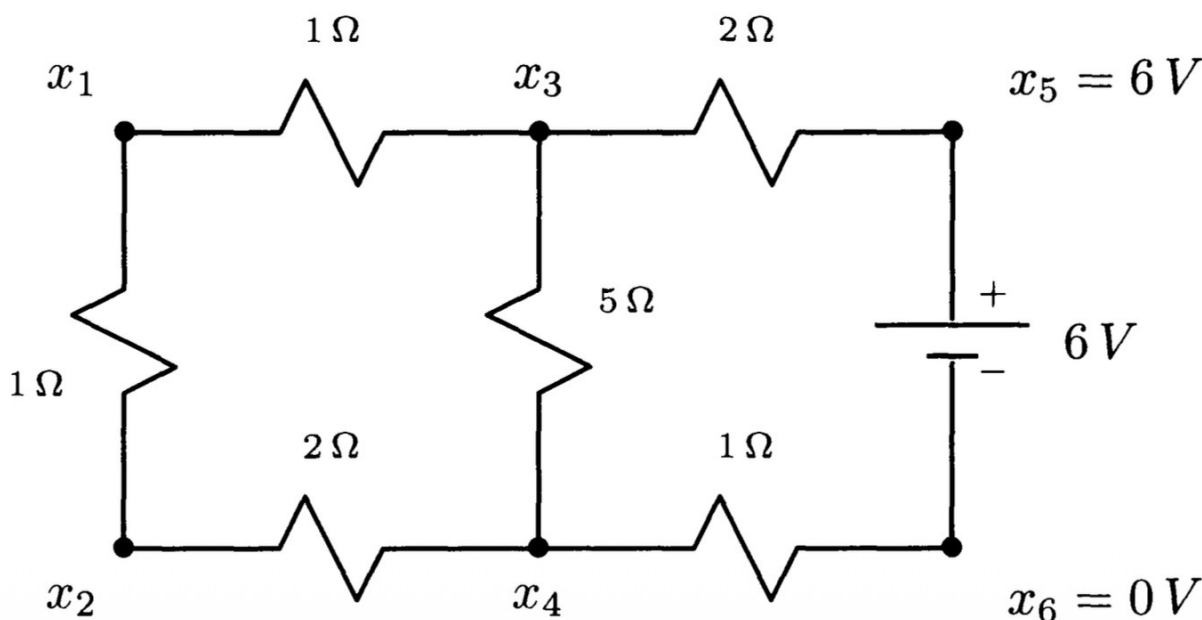


Figure 2: Electrical circuit

Exercise 4

Write your own Julia program that solves positive definite systems $\mathbf{A}\mathbf{x} = \mathbf{b}$ using subroutines to:

1. compute the Cholesky factor;
2. perform forward substitution;
3. perform back substitution.

Try your program on:

(i)

$$\begin{pmatrix} 36 & -30 & 24 \\ -30 & 34 & -26 \\ 24 & -26 & 21 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 12 \\ -7 \end{pmatrix}$$

(ii)

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 & 3 \\ 1 & 2 & 3 & 4 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 5 \\ 9 \\ 12 \\ 14 \\ 15 \end{pmatrix}$$

Exercise 5

Consider the differential equation

$$-u''(x) + cu'(x) + du(x) = f(x), \quad 0 < x < 1,$$

with boundary conditions:

$$u(0) = 0, \quad u(1) = 0.$$

We solve it using finite difference methods. Subdivide $[0, 1]$ into m subintervals of length $h = 1/m$. Let $x_j = jh$, $j = 0, \dots, m$ be grid points.

(a) Derive the discrete form:

$$\frac{-u_{i+1} + 2u_i - u_{i-1}}{h^2} + c \left(\frac{u_{i+1} - u_{i-1}}{2h} \right) + du_i = f_i$$

Apply boundary conditions $u_0 = u_m = 0$ to get $m - 1$ equations.

(i) Write the linear system $\mathbf{Ax} = \mathbf{b}$ for:

- (a) $m = 6$
- (b) $m = 8$
- (c) $m = 20$

Use $c = 1$, $d = 2$, and $f(x) = x^2 + 1$. Use ‘Tridiagonal’ in Julia to define $\mathbf{A}$.

(ii) Solve $\mathbf{Ax} = \mathbf{b}$ in each case and plot u against x in a single plot. Discuss and compare the curves.

Exercise 6

Linear Dynamical System: Owls and Mice

Consider $x_1(n)$ as the population of owls (in hundreds) at time n and $x_2(n)$ as the population of mice (in tens of thousands) at time n . Assume the following model:

$$\begin{aligned} x_1(n) &= \frac{2}{5}x_1(n-1) + \frac{3}{5}x_2(n-1) \\ x_2(n) &= -\frac{1}{10}x_1(n-1) + \frac{13}{10}x_2(n-1) \end{aligned}$$

Assume that at $n = 0$ we have $x_1 = 2$ and $x_2 = 3$ (this is 200 owls and 30,000 mice).

- (a) Represent this as the linear dynamical system $\mathbf{x}(n) = A\mathbf{x}(n-1)$. What is the matrix A ?
- (b) Determine the eigenvalues of A .
- (c) Find corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$.
- (d) Represent $\mathbf{x}_0 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ as $\mathbf{x}_0 = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2$.
- (e) Now use diagonalization to compute explicit expressions for $x_1(n)$ and $x_2(n)$ for any time n based on the expansion of $\mathbf{x}_0$ in the basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ as in (d).
- (f) Determine $\lim_{n \rightarrow \infty} \mathbf{x}(n)$. What is the meaning of this vector?
- (g) Plot the trajectory of $\mathbf{x}(n)$ (on the x_1, x_2 plane), and as functions of n (both plots for $x_1(n)$ and $x_2(n)$ on the same plot). Present the two alternative plots of this dynamical system, side by side. Make sure that the plots are neatly labeled and formatted. The plots should show convergence to the limiting point found in (f).
add the Plots package of julia by doing (using Pkg then `Pkg.add("Plots")`). You can learn more from Julia plotting here <https://docs.juliaplots.org/dev/tutorial/>